

DRISHTI

Predictive Crime Hotspot Mapping Assistant

IBM Bob AI Hackathon 2026

Track 4 • AI & Predictive

Turning historical crime records into explainable, time-aware patrol intelligence.

TEAM ANANTA

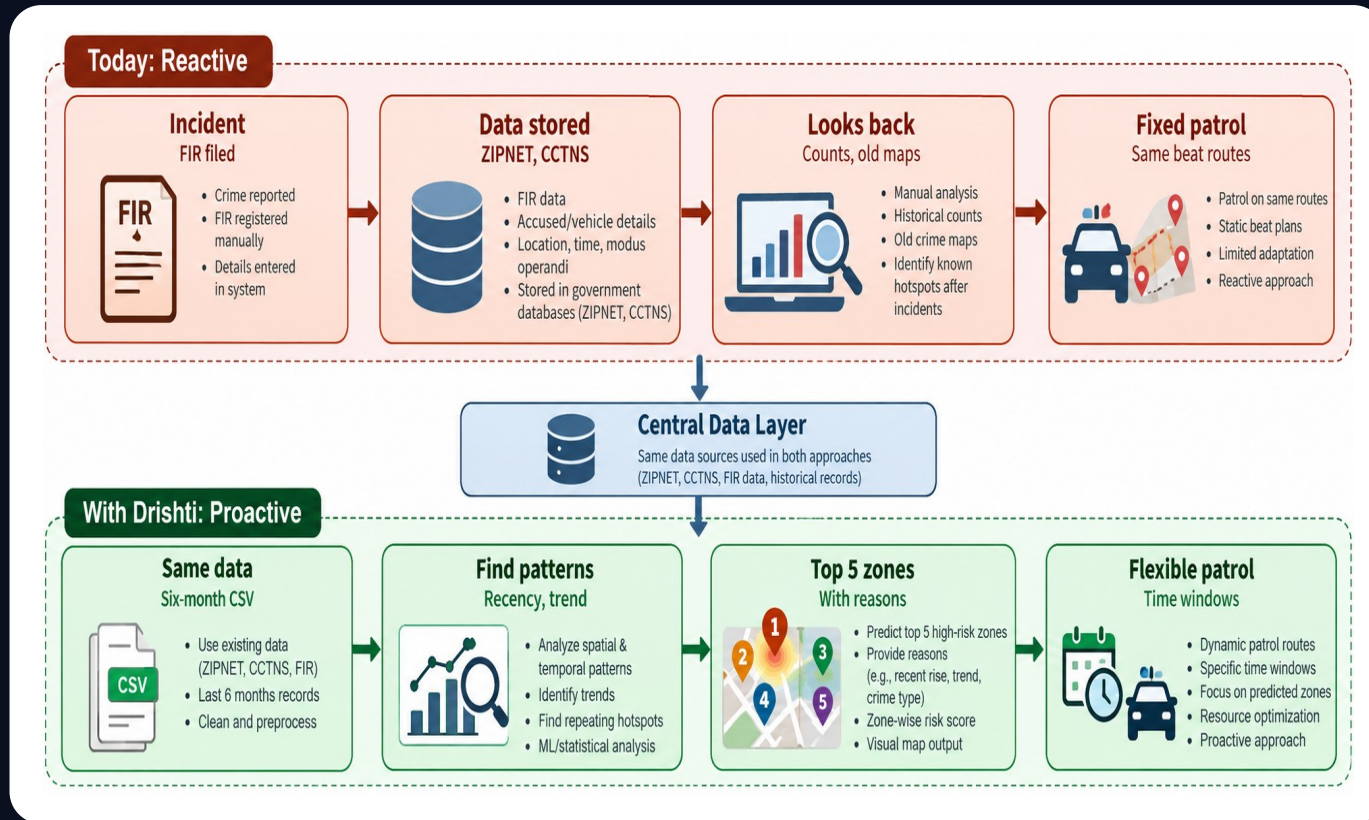
**Lakshay Sunil Jatawat • Kunj Buddhdev • Deeksha •
Jiya Joshi**

District-level decision support for proactive patrol planning



The gap is not data — it is predictive use of data

Police stations may already hold years of geo-tagged crime records, while patrol plans can remain static and reactive.



CURRENT STATE

Reactive workflow

- Incident is recorded after the event.
- Historical data is stored, but often looked at retrospectively.
- Beat routes can remain fixed despite changing spatial-temporal patterns.

Drishti adds a predictive decision layer on top of existing records.

Source basis: project README problem statement and solution description.

Drishti converts historical records into an actionable weekly brief

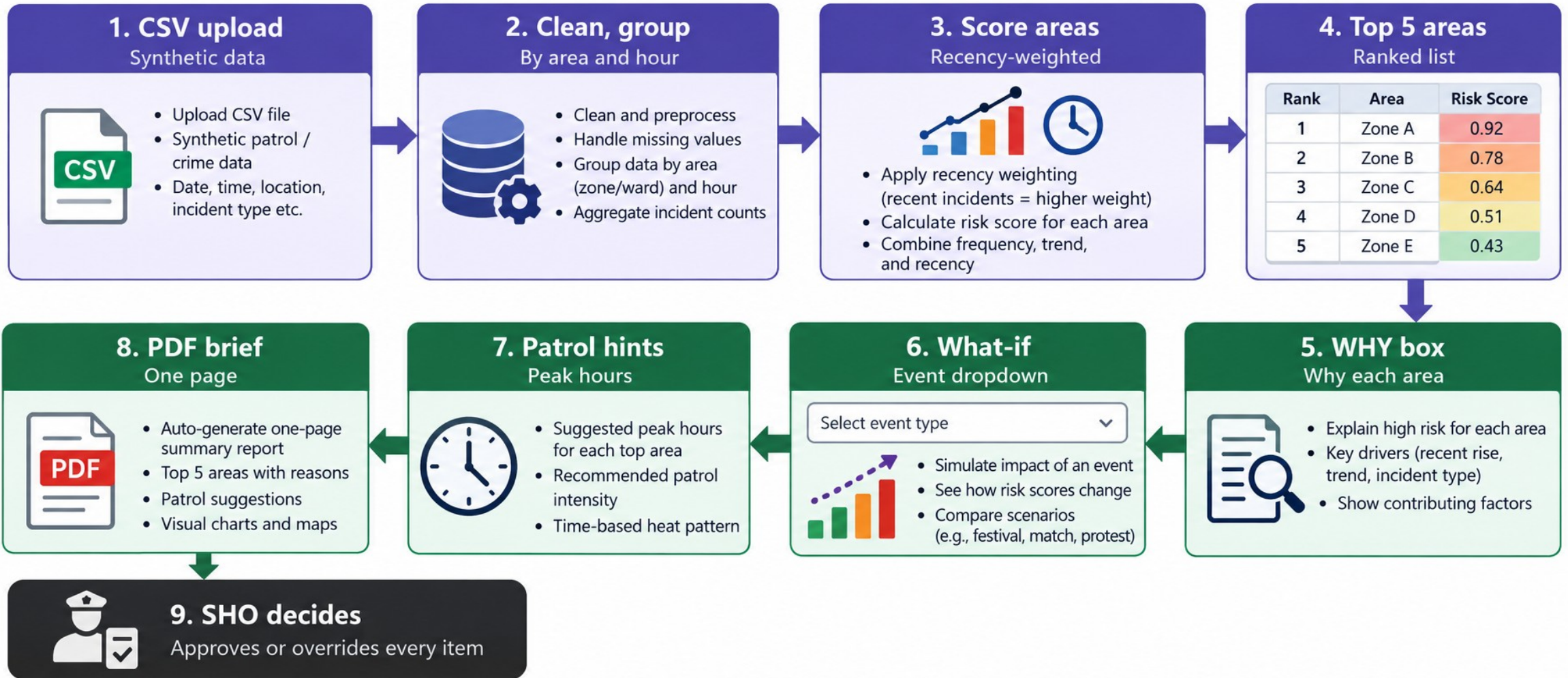
The system is designed around explainability and human approval — not black-box automation.



Key product promise:

Top-5 risk areas + the “why” + event impact + patrol timing — in one decision flow.

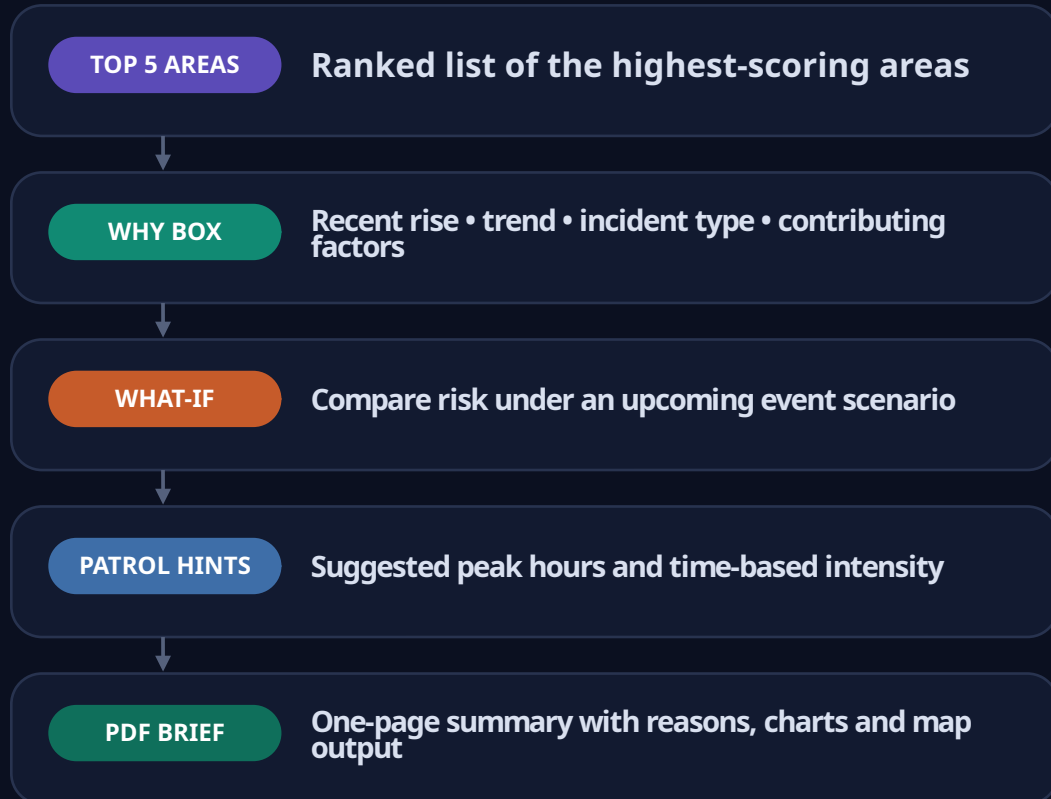
From CSV upload to an approved patrol decision



Human-in-the-loop: the SHO approves or overrides every recommendation.

The recommendation is explainable at every step

Drishti does not stop at a risk score; it exposes the reasoning and turns it into a usable operational brief.



OFFICER EXPERIENCE

What an SHO actually gets

- Readable** Plain-language explanation instead of only a model score.
- Scenario-aware** Event assumptions can be tested before deployment changes.
- Operational** Peak-hour guidance translates analysis into patrol timing.
- Controllable** Final approval stays with the SHO.

Decision support, not autonomous policing.

A lightweight stack built for a fast, explainable MVP

IBM Bob is the build accelerator; the application layer stays transparent and easy to inspect.

IBM TECHNOLOGY

IBM Bob

Used as the hackathon build environment / AI development layer.

Track 4: AI & Predictive

Rapid iteration + clear, auditable application logic

APP

Streamlit

Interactive district dashboard and workflow

DATA

Pandas • NumPy

Cleaning, grouping, aggregation and scoring inputs

VISUALIZATION

Plotly

Interactive maps / charts and scenario visuals

REPORTING

ReportLab • Matplotlib

One-page intelligence brief generation and charts

Repository modules include app.py, analytics.py, simulator.py, report_generator.py and generate_data.py.

What is implemented — and what is still a limitation

The current submission demonstrates the end-to-end intelligence workflow; it does not claim a field-validated forecasting model.

IMPLEMENTED

- 1000+ generated mock crime records for the MVP
- Spatial-temporal / near-repeat pattern analysis
- Recency-weighted top-5 area scoring
- Explainable “WHY” rationale for each area
- What-if event simulation
- Patrol hints + one-page PDF brief

LIMITATIONS

- Synthetic data** Uses generated mock records instead of real police records.
- Heuristic scoring** Weighted heuristic demo, not a calibrated probabilistic forecast.
- Grid mapping** Grid-based map rather than real GIS / H3 spatial indexing.
- Access control** SHO vs Beat Officer role separation is not implemented.

Next validation step: real data + calibrated forecasting + real GIS.

Team Ananta — building a decision layer for proactive patrol planning

The MVP is intentionally small: demonstrate the decision flow first, then harden it with real data and operational controls.

TEAM

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Team Lead

1 Kunj Buddhdev

2 Deeksha

3 Jiya Joshi

Built as Team Ananta for the IBM Bob AI Hackathon 2026.

NEXT STEP

From prototype → deployable intelligence layer

- Replace synthetic records with governed real police data
- Calibrate and validate the risk model on held-out / real-world data
- Move from grid maps to real GIS / H3 spatial indexing
- Add role-based access control and deployment hardening

Live demo is not deployed yet; repository includes demo artifacts.